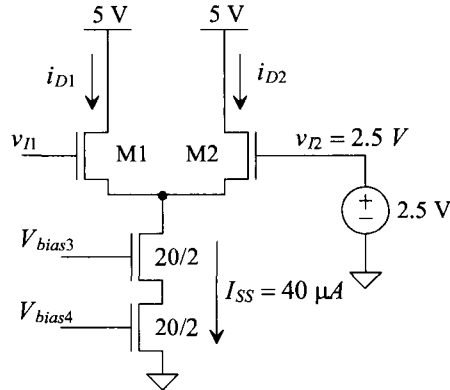




Parameters from Table 9.1
Bias circuit from Fig. 20.43

Figure 22.2 Diff-amp used in Ex. 22.1.

Using Eq. (22.7), we can estimate the maximum voltage on the gate of M1 as

$$v_{I1MAX} = \sqrt{\frac{2 \cdot L \cdot I_{SS}}{K P_n \cdot W}} + v_{I2} = \sqrt{\frac{2 \cdot 2 \cdot 40}{120 \cdot 10}} + 2.5 = 2.865 \text{ V}$$

or v_{I1} is 365 mV above v_{I2} . The minimum voltage allowed on the gate of M1 (to keep some current flowing in M1) is then 365 mV below v_{I2} or 2.135 V. The simulation results are seen in Fig. 22.3. Notice how, when the inputs are equal (both are 2.5 V), the currents in M1 and M2 are equal and approximately 20 μA ($= I_{SS}/2$). ■

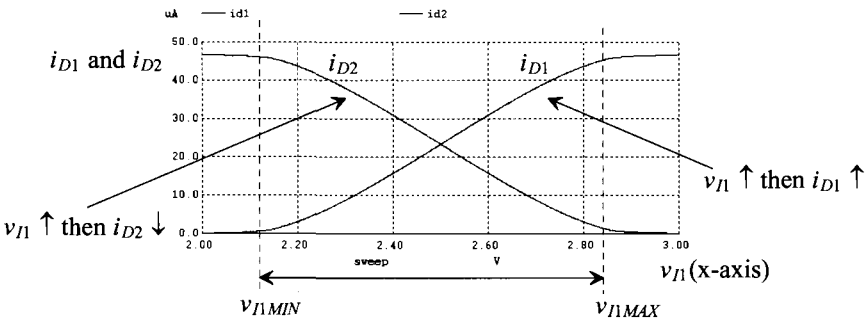


Figure 22.3 Simulating the operation of the diff-amp in Fig. 22.2.

Maximum and Minimum Common-Mode Input Voltage

When the diff-amp is used on the input of an op-amp, the inputs are forced, via feedback around the op-amp, to the same values (or very nearly the same values). This value (or more precisely the average of the two inputs) is called the *common-mode voltage*. Figure 22.4 shows the diff-amp with both inputs tied together. We're interested in the maximum and minimum voltage that will keep both M1 and M2 operating in the saturation region

(that is, not off or in the triode region). We'll call the maximum common-mode voltage, v_{CMMAX} and the minimum common-mode voltage, v_{CMMIN} . Note that when the common-mode voltage (the potential on the gates of M1 and M2) is too high, both M1 and M2 triode (behave like resistors). When the common-mode voltage is too low, M1 and M2 shut off.

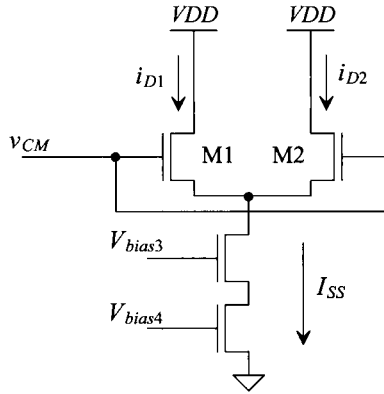


Figure 22.4 Diff-amp used to calculate the minimum and maximum diff-amp input voltages.

To determine the maximum input voltage, we know that for M1 and M2 to remain in the saturation region

$$V_{DS} \geq V_{GS} - V_{THN} \rightarrow V_D \geq V_G - V_{THN} \quad (22.9)$$

Because $V_D = VDD$, we can write

$$V_{CMMAX} = VDD + V_{THN} \quad (22.10)$$

The input common-mode voltage can actually be higher than VDD before M1/M2 move into the triode region.

For the minimum input voltage, we can write

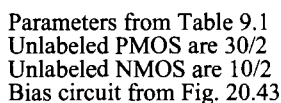
$$V_{CMMIN} = V_{GS1,2} + 2 \cdot V_{DS,sat} \quad (22.11)$$

where the minimum voltage across the current source is assumed to be $2V_{DS,sat}$. For the long-channel process, from Table 9.1, the minimum input voltage for an NMOS diff-amp is 1.55 V.

Example 22.2

Figure 22.5 shows a folded cascode amplifier (see Fig. 20.45). Assuming that all MOSFETs are operating in the saturation region, estimate the minimum and maximum input voltage of the amplifier.

Note how the widths of M5–M6 are doubled to sink the additional current from the diff-amp. It's important to understand how the sizes of the MOSFETs are adjusted in the current mirrors so that the currents sum correctly. The currents, as seen in Fig. 22.5, are labeled assuming $V_{plus} = V_{minus}$.



The maximum allowable input common-mode voltage is determined by

$$V_{CMMAX} = V_{DD} - V_{SG} - 2V_{SD,sat} = 5 - 1.15 - 0.5 = 3.35 \text{ V}$$

The minimum input common-mode voltage is determined by noting the drain voltage of M5 and M6 (or M1 and M2) is a $V_{DS,sat}$

$$V_{SD} \geq V_{SG} - V_{THP} \rightarrow V_D \leq V_G + V_{THP}$$

and so

$$V_{CMMIN} = V_D - V_{THP} = 0.25 - 0.9 = -0.65$$

In other words, the input common-mode voltage can actually be below ground and the amplifier will still function correctly. This result is *very useful*. The folded-cascode amplifier with PMOS diff-amp can be used when the input voltage swings around ground. Note, however, that the output voltage of the amplifier will not swing below ground but is, rather, limited to $2V_{DS,sat}$ above ground and $2V_{SD,sat}$ below V_{DD} if the MOSFETs are to remain in saturation (the high-gain region of operation). ■

Current Mirror Load

The previous uses of the diff-amp have exploited the fact that the output of the diff-amp is a current controlled by a differential input voltage, v_{Dr} . In some applications, it is desirable to have a diff-amp with a voltage output. Towards this goal, consider the diff-amp with a current mirror load, as seen in Fig. 22.6. An imbalance in the drain currents of M1 and M2 causes the output of the diff-amp to swing either towards V_{DD} or

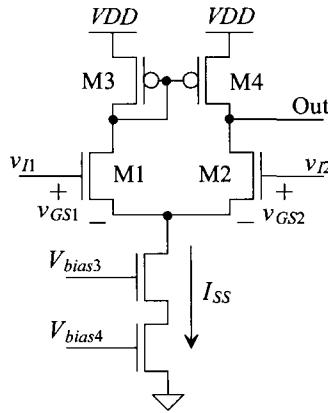


Figure 22.6 Diff-amp with a current mirror load.

ground. The minimum input common mode voltage is, once again, given by Eq. 22.11. The maximum input common mode voltage is determined knowing that the drain voltage of M2 is the same as the drain voltage of M1 (when both diff-amp inputs are the same potential), that is, $V_{DD} - V_{SG}$ of the PMOS. We can therefore write

$$V_{DS} \geq V_{GS} - V_{THN} \rightarrow V_D \geq V_G - V_{THN} \rightarrow V_{CMMAX} = V_{DD} - V_{SG} + V_{THN} \quad (22.12)$$

For the parameters in Table 9.1, $V_{CMMAX} = 5 - 1.15 + 0.8 = 4.65 \text{ V}$.

The voltage output swing can be determined by noting that the maximum voltage output is limited by keeping M4 in saturation. Therefore,

$$V_{OUTMAX} = V_{DD} - V_{SD,sat} \quad (22.13)$$

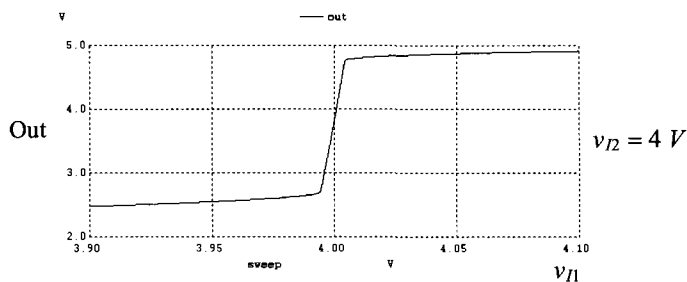
The minimum output voltage is determined by the voltage on the gate of M2 (M2 must remain in saturation).

$$V_D \geq V_G - V_{THN} \rightarrow V_{OUTMIN} = V_{G2} - V_{THN} \quad (22.14)$$

Example 22.3

Suppose the diff-amp in Fig. 22.6 is implemented with the sizes and bias currents seen in Table 9.1 ($I_{SS} = 40 \mu\text{A}$). If the gate of M2 is held at 4 V, estimate the diff-amp's output swing. Verify the answers with SPICE.

From Eq. (22.13), the output can swing up to 4.75 V before M4 triodes. From Eq. (22.14), the output can swing down to $4 - 0.8$ or 3.2 V before M2 triodes. The simulation results are seen in Fig. 22.7. Looking at the results, we see the output goes down to approximately 2.8 V or 400 mV less than what we predicted. This can be attributed to the body effect in M1 and M2. With body effect, the threshold voltage is 1.2 V (instead of 0.8 V). ■



Biasing from the Current Mirror Load

Looking at this equation, we see that the V_{SG} of M3 is the “weak link” in the minimum power supply voltage. Using the parameters from Table 9.2 (the short-channel devices with a V_{DD} of 1 V), we get a $V_{DD_{min}}$ of 500 mV. We can lower this value by using a single MOSFET to bias the diff-amp (instead of a cascode circuit).

22.1.2 AC Operation

To describe the AC small-signal operation of the diff-amp in Fig. 22.1, consider the AC schematic seen in Fig. 22.9. We've replaced the current source with an open and the DC supply (VDD) with a short (to ground). In the following discussion, it's important to remember that a negative AC current simply means that the overall current (AC + DC) is decreasing (see Ex. 9.5 in Ch. 9). We can write

$$v_{di} = v_{i1} - v_{i2} = v_{gs1} - v_{gs2} = \frac{i_{d1}}{g_m} - \frac{i_{d2}}{g_m} \quad (22.16)$$

Reviewing Fig. 22.9, we see that

$$i_{d1} = -i_{d2} = i_d \quad (22.17)$$

and thus

$$v_{gs1} = -v_{gs2} = \frac{v_{di}}{2} \quad (22.18)$$

so finally

$$g_m v_{gs1} = g_m (-v_{gs2}) = i_d = g_m \cdot \frac{v_{di}}{2} \quad (22.19)$$

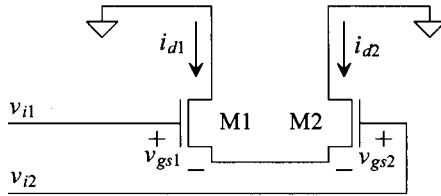


Figure 22.9 AC circuit for a diff-amp.

Example 22.4

Estimate the AC components of the drain currents of M1 and M2 for the circuit in Fig. 22.2 if $v_{i1} = 2.5 + 1\text{mV} \cdot \sin(2\pi \cdot 1\text{kHz} \cdot t)$. Verify your hand calculations using SPICE.

From Table 9.1 we know the g_m of the NMOS devices used in the diff-pair is $150 \mu\text{A/V}$. The AC component of v_{i2} is zero so we know

$$v_{gs1} = -v_{gs2} = 0.5\text{mV}$$

and thus

$$i_d = g_m v_{gs} = (150 \times 10^{-6})(500 \times 10^{-6}) = 75\text{ nA}$$

This is the AC component of the drain currents. When $i_{d1} = 75\text{ nA}$, then $i_{d2} = -75\text{ nA}$. The overall drain currents are written as

$$i_{D1} = 20\text{ }\mu\text{A} + (75\text{ nA}) \cdot \sin(2\pi 1\text{kHz} \cdot t) \text{ and } i_{D2} = 20\text{ }\mu\text{A} - (75\text{ nA}) \cdot \sin(2\pi 1\text{kHz} \cdot t)$$

The simulation results are seen in Fig. 22.10. ■

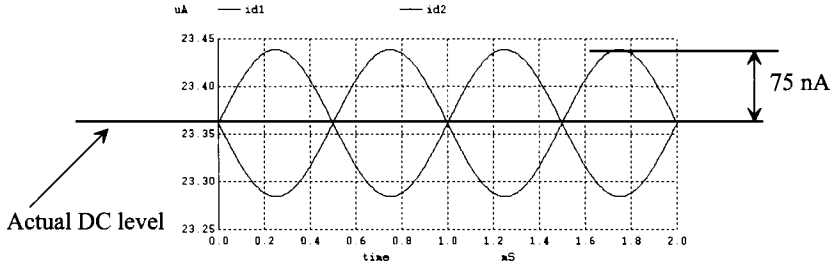


Figure 22.10 Simulation results for Ex. 22.4.

AC Gain with a Current Mirror Load

To determine the AC gain of the diff-amp with a current mirror load, Fig. 22.6, consider the small signal model seen in Fig. 22.11. We've replaced the diode-connected MOSFET, M3, with a $1/g_{m3}$ resistor. Also the resistance looking into the drain of M4 (r_{o4}) and the resistance looking into the drain of M2 (r_{o2}) are drawn explicitly. We assumed that the small-signal resistance looking into the drain of M2 is simply r_{o2} (for all practical design cases this is a good assumption). Note how, because of the current mirror action, the AC current flowing in M4 is i_{d1} . The current in M3 is mirrored by M4. The output voltage can be written as

$$v_{out} = (i_{d1} - i_{d2}) \cdot (r_{o2} \parallel r_{o4}) \quad (22.20)$$

Knowing $i_{d1} = -i_{d2} = i_d$

$$v_{out} = 2i_d \cdot (r_{o2} \parallel r_{o4}) \quad (22.21)$$

The differential mode gain is then written with the help of Eq. (22.19) as

$$A_d = \frac{v_{out}}{v_{di}} = \frac{v_{out}}{v_{i1} - v_{i2}} = g_m \cdot (r_{o2} \parallel r_{o4}) \quad (22.22)$$

Notice that as the voltage on the gate of M1 increases, the current in M1 (and M3/M4) increases. This causes the output voltage to increase. When the gate potential of M2 increases, so does its drain current, causing the output voltage to decrease. Sometimes the gate potential of M1 is called the *noninverting* input and the gate potential of M2 is called the *inverting* input.

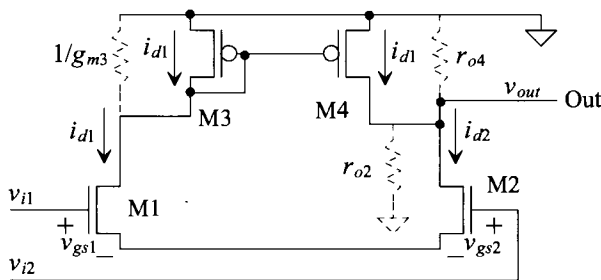


Figure 22.11 AC circuit of the diff-amp with a current-source load.

Example 22.5

Determine the output voltage for the diff-amp seen in Fig. 22.12. Verify your hand calculations using SPICE.

The tail current of the diff-amp conducts $20\text{ }\mu\text{A}$ so M1 and M2, when the inputs are equal, each conduct $10\text{ }\mu\text{A}$. Using the parameters from Table 9.2 and Eq. (22.22), the gain of the diff-amp is

$$A_d = g_m \cdot (r_{on} \parallel r_{op}) = (150 \times 10^{-6}) \cdot \left(\frac{167 \cdot 333}{167 + 333} \times 10^3 \right) = 16.7\text{ V/V}$$

This means that the 1 mV AC input will appear as a 16.7 mV AC output. The DC level on the output is $V_{DD} - V_{SG} = 650\text{ mV}$. The output voltage is then

$$v_{out}(t) = 0.65 + (16.7\text{ mV}) \cdot \sin(2\pi 10\text{ MHz} \cdot t)$$

The simulation results are seen in Fig. 22.13. ■

Bias circuit from Fig. 20.47.
See Table 9.2.

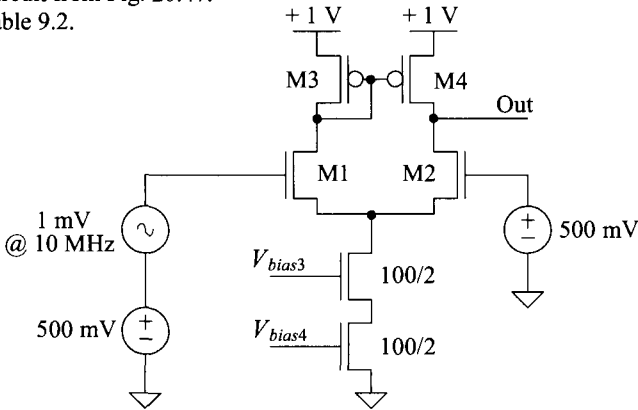


Figure 22.12 Circuit used in Ex. 22.5.

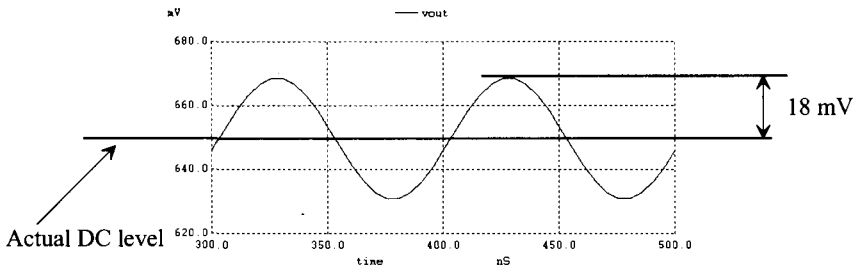


Figure 22.13 The output of the circuit in Fig. 22.12. Note that the transient analysis doesn't show the beginning of the simulation where the reference circuit is starting up.

Example 22.6

Estimate the f_{3dB} of the diff-amp in Ex. 22.5 if it drives a load capacitance of 1 pF. Verify your hand calculations with SPICE.

The resistance on the output node is $r_{o2} || r_{o4} = 111 \text{ k}\Omega$; therefore, since this is the *high-impedance node*, we can estimate the circuit's 3-dB frequency in the circuit as

$$f_{3dB} = \frac{1}{2\pi \cdot 111 \text{ k} \cdot 1 \text{ pF}} = 1.4 \text{ MHz}$$

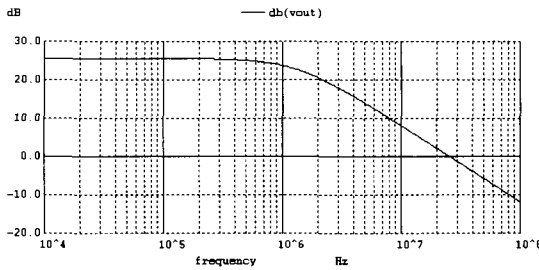


Figure 22.14 The AC response of the diff-amp in Fig. 22.12 when driving a 1 pF load capacitance.

The simulation results are seen in Fig. 22.14. ■

22.1.3 Common-Mode Rejection Ratio

An important aspect of the differential amplifier is its ability to reject a common signal applied to both inputs. Often, in analog systems, signals are transmitted differentially, and the ability of an amplifier to reject coupled noise into each line is very desirable. Consider the amplifier shown in Fig. 22.15. The bias current at the sources of M1/M2 has been replaced with its small-signal output resistance. The common-mode signal (the “noise” labeled in the figure) on both inputs is, ideally, rejected by the diff-amp. The diff-amp’s output voltage doesn’t vary. The noise causes the source potentials of M1/M2 to vary (the voltage across the tail current source).

If we apply an AC signal, v_c , to the gates of M1/M2, equivalent to saying that a common signal is applied to the input of the differential amplifier, we can calculate the common-mode gain. We begin by writing the AC small-signal, common-mode input voltage, v_c , as

$$v_c = v_{gs1,2} + 2i_d R_o \quad (22.23)$$

The MOSFETs M1 and M2 each source a current, i_d , through the output resistance of the current source, R_o (hence the factor of two in this equation). This equation may be rewritten as

$$v_c = i_d \left(\frac{1}{g_m} + 2R_o \right) \approx i_d 2R_o \quad (22.24)$$

The output voltage, because of the symmetry of the circuit, is given by

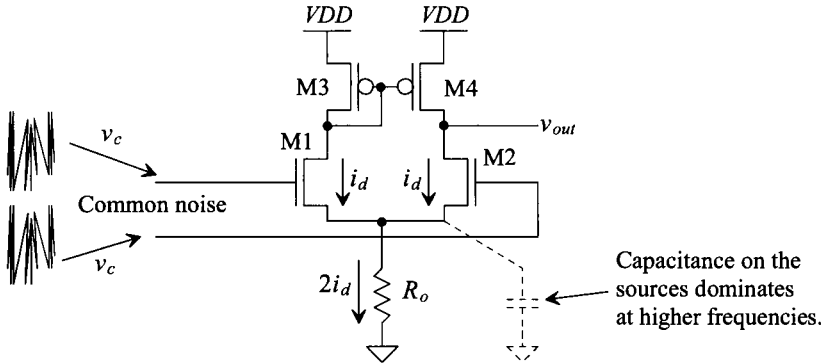


Figure 22.15 Calculating the CMRR of a diff-amp.

$$v_{out} = -i_d \cdot \frac{1}{g_{m3}} = -i_d \cdot \frac{1}{g_{m4}} \quad (22.25)$$

The common-mode gain is then

$$A_c = \frac{v_{out}}{v_c} = \frac{-1/g_{m3,4}}{2R_o} = -\frac{1}{2g_{m3,4}R_o} \quad (22.26)$$

Notice that by increasing the output resistance of the biasing current source, the common-mode gain drops. (This is good because we don't want any common signal to be amplified by the diff-amp.) The common-mode rejection ratio (*CMRR*), in dB, of a diff-amp with a current mirror load is given by

$$CMRR = 20 \cdot \log \left| \frac{A_d}{A_c} \right| = 20 \cdot \log [g_{m1,2}(r_{o2} || r_{o4}) \cdot 2g_{m3,4}R_o] \quad (22.27)$$

The larger the *CMRR*, the better the performance of the diff-amp. For the diff-amp in Fig. 22.12, Eq. (22.27) evaluates to 86 dB. As the frequency of the input common-mode signal, v_c , increases, the *CMRR* goes down. This is the result of the impedance of the capacitance to ground on the sources of the diff-pair becoming much smaller than R_o .

Example 22.7

Simulate the *CMRR* of the diff-amp in Fig. 22.12. Show that the *CMRR* falls with increasing frequency.

As mentioned above, the (low-frequency) calculated *CMRR* is 86 dB. To simulate the *CMRR*, we perform AC simulations on two of the diff-amps in Fig. 22.12 so we can simultaneously determine A_d and A_c . The ratio of these gains is the *CMRR*. Figure 22.16 shows the simulation results. The common-mode gain, A_c , depends on the DC common-mode voltage. The results seen in Fig. 22.16 are with a common-mode (DC) voltage of 700 mV. Dropping the common-mode voltage down to 500 mV causes the low-frequency *CMRR* to drop down to 50 dB. The reason for this is that, with higher DC common-mode voltages, the voltage across the tail current source is higher, so its output resistance is larger. ■

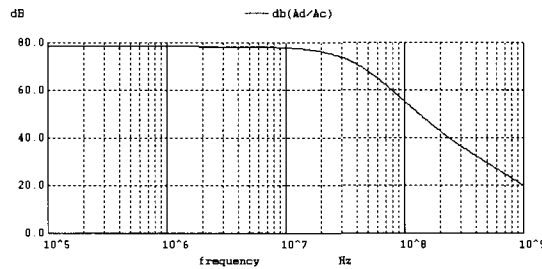


Figure 22.16 The simulated CMRR for the diff-amp in Fig. 22.12.

Input-Referred Offset from Finite CMRR

It's important we understand, in Eq. (22.26), why large R_o reduces A_c . Looking at Fig. 22.15, notice that as v_c varies so too will the voltage across the biasing current source (represented in this figure as R_o). As seen in Ch. 20, Fig. 20.4, variations in the voltage across a current source changes the output of the current source (unless its output resistance is infinite). This variation in current causes the current in M1 and M2 to change with variations in v_c . Changes in the drain current of M1/M2 (and thus M3/M4) causes the output voltage to change. Figure 22.17 shows how the output voltage changes as a function of the DC common-mode voltage. Understanding this circuit is *very important*. If we wanted to hold the diff-amp output voltage constant while varying the common mode voltage, we would have to apply a small voltage difference between the gates of M1 and M2 (*an offset voltage*). We'll revisit this important topic when we discuss op-amp CMRR.

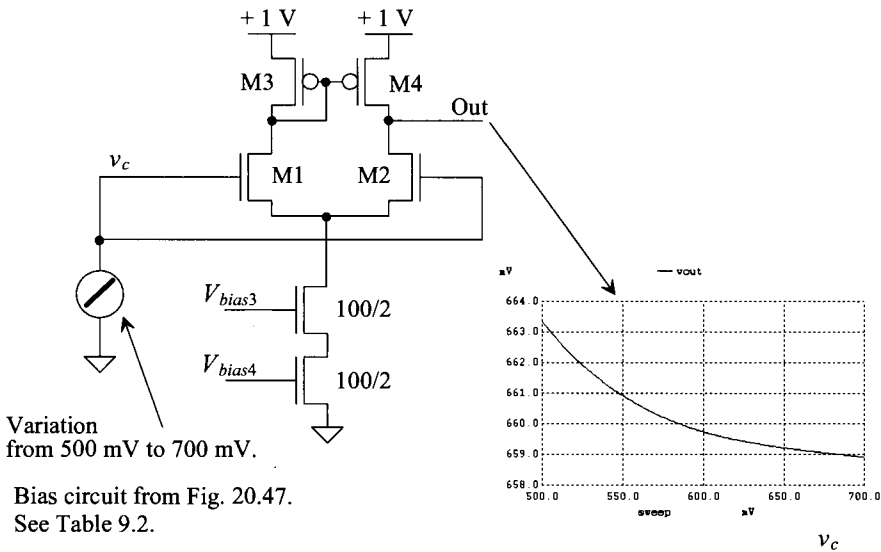


Figure 22.17 How variation in common-mode voltage affects the output voltage.